

Multiple Choice (10 marks)

1. What drives differentiation?
 - (a) Spontaneous emission from radioactive atoms.
 - (b) The minimization of gravitational potential energy.
 - (c) Thermally induced collisions.
 - (d) Plate tectonics.
2. Why do we see essentially the same face of the Moon at all times?
 - (a) because the Moon has a nearly circular orbit around the Earth
 - (b) because the Moon does not rotate
 - (c) because the other face points toward us only at new moon when we can't see the Moon
 - (d) because the Moon's rotational and orbital periods are equal
3. Previous IAAC rounds featured Proxima/Alpha Centauri as closest star(system) to the Earth. Which one is the second closest star(system)?
 - (a) Wolf 359
 - (b) Sirius
 - (c) 61 Cygni
 - (d) Barnard's Star
4. Why is the sky blue?
 - (a) Because the molecules that compose the Earth's atmosphere have a blue-ish color.
 - (b) Because the sky reflects the color of the Earth's oceans.
 - (c) Because the atmosphere preferentially scatters short wavelengths.
 - (d) Because the Earth's atmosphere preferentially absorbs all other colors.
5. Which of the following characteristics would not necessarily suggest that a rock we found is a meteorite.

- (a) It has a fusion crust
- (b) It contains solidified spherical droplets
- (c) It is highly processed
- (d) It has different elemental composition than earth

True / False (10 marks)

Answer True or False

- 6. True or False: The Sun mainly emits blue light, which causes the sky to appear blue during the day.
- 7. True or False: The surface of Venus is approximately 2 billion years old, suggesting a much older geological formation than is accurate.
- 8. True or False: Venus lacks seasons because its rotation axis is nearly perpendicular to the plane of the Solar System.
- 9. True or False: Bolometric luminosity in astronomy is the luminosity integrated only over visible wavelengths of light.
- 10. True or False: Saturn's equatorial wind speed is not greater than 900 miles per hour, unlike Jupiter's high-speed winds.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the implications of observing a hydrogen spectral line at 485.9 nm in a star's spectrum, compared to the known laboratory wavelength of 486.1 nm. What does this indicate about the star's motion?
- 12. Discuss the categories of electromagnetic radiation, analyzing their wavelengths from shortest to longest, and explain the implications of this order for astronomical observations.

End of Paper